Memduh Burak Ardıç

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Passport: [redacted] **Gender:** Male **Date of birth:** 05/06/1995 **Place of birth:** Eregli, Türkiye **Nationality:** Turkish

ABOUT ME

I'm a Water Resources Engineer who loves solving problems, especially in modeling. My interests include environmental modeling, numerical modeling, and climate scenario analysis.
My current research focuses on applying integrated modeling systems to evaluate how climate change impacts hydrological systems, water bodies, and environmental dynamics. I have experience working with observational datasets, spatial data, and global climate model outputs to prepare model simulations.
Professionally, I love contributing to projects that support sustainable water management, environmental risk assessment, and climate adaptation strategies. I also have a growing affinity and enthusiasm for remote sensing and Earth observation technologies, especially in the applications of hydrology and environmental problems, as well as supporting decision-making in water resources management.

EDUCATION AND TRAINING

[03/09/2018 – 15/09/2021]

MSc., Civil Engineering (Hydraulics and Water Resources Management)

Gazi University https://gazi.edu.tr/

City: Ankara | **Country:** Türkiye | **Field(s) of study:** Engineering, manufacturing and construction | **Final grade:** 3.61/4 | **Thesis:** Hydrodynamic and water quality modeling in the tidal estuaries: The Ribble Estuary case

Focused on computational fluid dynamics in environmental hydraulics. Especially in water quality and sediment transport modeling

[02/09/2013 – 18/06/2018]

BSc., Civil Engineering

Gazi University https://gazi.edu.tr/

Country: Türkiye | **Field(s) of study:** Engineering, manufacturing and construction | **Thesis:** Numerical Analysis of Seepage and Flow Nets by Finite Differences Method

WORK EXPERIENCE

NFB Engineering and Consulting Inc.

City: Ankara | **Country:** Türkiye

[01/03/2023 – Current]

Expert

Project: Technical Assistance on Preparation of River Basin Management Plans for Six Basins Project-funded by EU and Türkiye Goverment.

https://6rbmp.tarimorman.gov.tr/en

- Contributing to the basin-wide modeling framework, including river network topology development, hydrological simulation, water allocation modeling, and water quality modeling. Responsibilities include model setup, configuration, calibration, and scenario definition using WEAP and AquaTool (EVALHID–SIMGES–GESCAL).
- Integrating climate change impacts into modeling workflows by evaluating multiple Global Climate Models, applying statistical and dynamical downscaling approaches, generating bias-corrected climate forcings, and conducting trend and scenario analyses to assess future hydrological and water quality conditions.
- Leading drought characterization and risk assessment activities, including computation of hydrologic drought indices, spatiotemporal drought analysis, and evaluation of the interactions between low-flow conditions and water quality deterioration within the basin systems.
- Supporting inter-model coupling and data harmonization, ensuring consistency between hydrological, water allocation, and water quality modules, managing time-series and spatial data preprocessing, and coordinating input–output workflows across modeling platforms.

- Delivering technical presentations and model result briefings to clients, regulatory bodies, and stakeholder groups as part of collaborative basin planning and review processes.
- Participating in field investigations to support data collection, ground-truthing, stakeholder engagement, and validation of model assumptions and boundary conditions.
- Working with remote sensing and Earth observation datasets, such as Sentinel-1/2, ERA5, and MODIS, to support hydrological and environmental assessments through spatial analysis, watershed characterization, extraction of environmental parameters for model inputs and scenario evaluation, and integrating these datasets into machine learning workflows for classification and predictive modeling tasks.

NFB Engineering and Consulting Inc.

City: Ankara | **Country:** Türkiye

[24/03/2021 – 01/03/2023] **Project Engineer**

Project: Preparation of Yesilirmak, Bati Karadeniz, Sakarya, and Susurluk Basins Drought Management Plans

- Coordinated the preparation of the technical reports on the drought management plans which include water budget calculations, drought analysis, sectoral vulnerability analysis, and measure plans.
- Managed, processed, and visualized the hydrological and meteorological data for 4 different basins by using the GIS software (ArcGIS, QGIS), spreadsheet programs, and scripts in C++, Python, and R languages.
- Set-up 2 different hydrological models (GR2M and Hec-HMS) to determine predicted water potential under the Global Climate Model outputs.
- Took part in drought analysis for future periods prediction works with Global Climate Model outputs (Calculations of several drought indices (such as SPI, SPEI, Palmer Drought Indices, etc.) on basin-scale)
- Led the sectoral vulnerability analysis for all basins which was performed in agriculture, industry, drinking and utility water, ecosystem, tourism, energy, and health sectors to determine the impacts of drought in each sector.
- Led the specifying measures in drought mitigation for all basins.
- Took a role in a training program about hydrological and hydrodynamical modeling.
- Gave one lecture about the application of HEC-HMS at Yıldırım Beyazıt University to undergraduate students.
- Made presentations about the prepared technical reports to the clients and stakeholders.

Gazi University

City: Ankara | **Country:** Türkiye

[03/09/2018 – 01/10/2021] **Researcher-Graduate Student**

- Prepared 3-dimensional hydrodynamical models to simulate the tidal forces in the Ribble Estuary by using the Delft3D, DIVAST, and SWAN and determined wet-dry areas and hydrodynamical quantities
- Researched the topic of sediment transport by using and developing numerical models
- Tested the performance and efficiency of the different numerical methods, grid generation techniques, and turbulence modeling by running prepared hydrodynamic models
- Wrote scripts that have Navier-Stokes equation and Advective Diffusion equation solvers in Python and FORTRAN languages to use in the research projects
- Prepared the water quality models to predict the distribution of Fecal Coliform concentration discharged to the Ribble Estuary by using D-Water Quality and the outputs of the hydrodynamical models

io Environmental Solutions Research and Development Co.

City: Ankara | **Country:** Türkiye

[24/09/2018 – 01/10/2019] **Project Engineer**

- Worked on the hydrological modeling in 3 different basins by setting up hydrological and meteorological data in the models.
- Assisted the water budget works in 3 different basins by processing the collected data from the field
- Took part in technical reporting.

LANGUAGE SKILLS

Mother tongue(s): Turkish

Other language(s):

English

LISTENING B2 **READING** B2 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Python | FORTRAN | MATLAB | Environmental Modeling | Water Quality Modeling | Hydrological Modeling | SWAT | Delft3D | HEC-HMS | HBV | ArcGIS | QGIS | Microsoft Office | Project Management | GIS | Remote Sensing

PUBLICATIONS

[2026] **Developing a comparative decision support framework through AQUATOOL and WEAP integration: Case of Manavgat Basin.**

Authors: Kacikoc, M., Mesta, B., Ardic, M. B., Yalcin, A., & Turan, B. | **Journal Name**: Under review.

[2021] **Investigation of a Type of the Local Grid Refinement in the Model Harbour**

Reference: Koçyiğit, Ö. and Ardıç, M. B. , "Investigation of a Type of the Local Grid Refinement in the Model Harbour," Middle East International Conference On Contemporary Scientific Studies-V , vol.1, no.1, Ankara, Turkey, pp.548-557, 2021
Journal Name: Middle East International Conference On Contemporary Scientific Studies-V | **Volume, Issue and Pages**: vol.1, no.1, Ankara, Turkey, pp.548-557

CONFERENCES AND SEMINARS

[05/09/2024 – 06/09/2024] **Training Workshops on Modeling, Environmental Objectives and Programme of Measures I**

Ankara

I presented on climate change within the context of modeling studies.

[26/11/2024 – 27/11/2024] **Improving Public and Sector Representatives's Awarness on Water Allocation via Traning Programs, Workshops, and Informative Meeting**

Ankara

I presented water allocation strategies during drought periods as part of the project.

[30/01/2025 – 31/01/2025] **Training Workshops on Modeling, Environmental Objectives and Programme of Measures II**

Ankara

I gave a presentation about climate change approaches, GCMs, and RCMs.

[12/07/2019 – 13/07/2019] **Methodologies and Applications in Project Management** Ankara

CERTIFICATIONS

[British Council, 07/04/2025] **IELTS Academic (Band 6.5)**

Mode of learning: Presential

[Project Management Institute, 14/07/2019] **Methodologies and Applications in Project Management**

Mode of learning: Presential

RECOMMENDATIONS

Name: Dr. Onder Kocyigit Supervisor

Senior Lecturer at Civil Engineering Department Faculty of Engineering Gazi University

Phone number: (+90) 3125823212 | **Email:** konder@gazi.edu.tr

Name: Assoc. Prof. Musteyde Baduna Kocyigit Supervisor

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